

NASCAR Sponsorship ROI Analysis

A Visibility Scoring and Efficiency Study of the 2024 Cup Series

Prepared for: NY Racing

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Scope: 4 primary sponsors, 36 races, 144 observations

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Executive Summary

NY Racing is evaluating NASCAR Cup Series sponsorship opportunities and needs an objective way to compare partners. Historically these decisions rest on relationships and intuition rather than quantitative visibility data, so the team cannot compare opportunities fairly or identify which partnerships deliver the best return. This analysis answers a specific question: **which sponsors deliver the best combination of visibility and value, and which should NY Racing prioritize?**

Our answer: FedEx is the balanced best bet. Across four primary sponsors and 36 races of the 2024 season, FedEx ranks #2 in both total visibility and cost efficiency, the best combined position of any sponsor. It captures 99.7% of the maximum exposure at a materially better value than the top-visibility option. For teams prioritizing ROI, Love's Travel Stops offers the best value per dollar; for teams prioritizing raw exposure, NAPA Auto Parts delivers the most.

Scope

- **Sponsors analyzed:** 4 (FedEx, NAPA Auto Parts, McDonald's, Love's Travel Stops)
- **Season:** 2024 NASCAR Cup Series, 36 races (144 sponsor-race observations)

- **Methodology:** a composite 0-100 visibility score combining race performance (40%), media coverage (30%), social engagement (20%), and special events (10%), divided by estimated cost to measure efficiency.

Headline Findings

Metric	Leader	Value
Highest visibility	NAPA Auto Parts	1,219 points
Best efficiency	Love's Travel Stops	82.2 points per \$M
Best balance (recommended)	FedEx	#2 visibility, #2 efficiency

1. **Efficiency flips the ranking.** The highest-visibility sponsor (NAPA) is the least efficient; the lowest-visibility sponsor (Love's) is the most efficient. Raw exposure and value per dollar point to different winners.
2. **On-track performance drives visibility only weakly in this data.** The one statistically significant performance-to-visibility link was laps led to news coverage ($r = +0.181$, $p = 0.030$); finish position showed almost no linear relationship with exposure.
3. **The rankings are robust.** Under four different weight configurations the ranking moved at most one position (only the top two swap), so the conclusions do not hinge on the exact weights.

Recommendation

FedEx as the primary target (balanced exposure and value), with **Love's** as the value fallback if budget tightens and **NAPA** as the premium option if maximum exposure is the goal.

Most Important Caveat

Sponsorship costs are confidential; our figures are benchmark estimates. Efficiency rankings were tested against low, mid, and high cost estimates and the core conclusions held. See Section 4 (Limitations) for the full discussion.

1. Methodology

1.1 Data Sources

The analysis draws on four data categories collected for the 2024 NASCAR Cup Series. Each was verified for coverage and quality before use.

Category	Source	Collection Method	Notes
Race performance	Kaggle NASCAR dataset, Racing Reference	Download + manual verification	Finish position, laps led; no missing races for tracked sponsors
Social (Reddit)	Google Trends proxy	Trends API	Reddit's public API returned HTTP 403, so Google Trends interest was substituted; race-level
Social (YouTube)	YouTube Data API	API	Sponsor-relevant video views; sparse (non-zero in 4 of 144 rows)
Media (news)	Manual search, tiered weighting	Search + classification	Tier 1 (ESPN, NASCAR.com) = 3, Tier 2 (regional) = 2, Tier 3 (blogs) = 1; sponsor-specific
Cost estimates	Forbes, Sports Business Journal, tier benchmarks	Estimates	Exact figures confidential; low / mid / high ranges recorded

Data quality notes. The Kaggle race data was cross-checked against Racing Reference; sponsor identification was added manually from team livery data. Social and news data were collected within a consistent time window across all sponsors so the comparison is fair. The most important quality caveat is that Reddit (a Google Trends proxy) and total YouTube views are *race-level* signals - identical for all four sponsors within a race - and therefore cannot distinguish sponsors on their own. Only news mentions and sponsor-specific YouTube views vary at the sponsor level.

1.2 Sponsor Selection

Four primary sponsors were analyzed. Selection criteria were: full-season primary sponsorship, sufficient data across all four channels, and a deliberate mix of team tiers so the model is tested against both elite and mid-pack operations.

Sponsor	Team	Driver	Team Tier
FedEx	Joe Gibbs Racing	Denny Hamlin	Top
NAPA Auto Parts	Hendrick Motorsports	Chase Elliott	Top
McDonald's	23XI Racing	Bubba Wallace	Mid-to-top
Love's Travel Stops	Front Row Motorsports	Michael McDowell	Mid

1.3 Visibility Scoring Model

We built a composite visibility score that combines four categories into a single 0-100 metric. Category weights start from industry sponsorship-ROI frameworks and are refined by the exploratory analysis. News carries the highest single weight because it was the only sponsor-varying exposure channel with a statistically significant link to on-track performance.

Category	Variable	Effective Weight	Transform
Race Performance	finish_position	28%	invert_position
Race Performance	laps_led	12%	normalize
Media Coverage	news_weighted_mentions	30%	normalize
Social Engagement	reddit_mentions	10%	normalize
Social Engagement	youtube_sponsor_views	10%	log_normalize
Special Events	is_win	6%	binary
Special Events	is_playoff	4%	binary

Normalization. All variables are scaled to a 0-100 range with min-max normalization computed season-wide (across all 144 rows) so every sponsor sits on one shared scale.

- Direct variables (higher is better): $\text{score} = (\text{value} - \text{min}) / (\text{max} - \text{min}) * 100$.
- Inverse variables (finish position, lower is better): $\text{score} = (\text{max} - \text{value}) / (\text{max} - \text{min}) * 100$, so a race win (P1) maps to 100.
- YouTube views are log-transformed ($\log(x + 1)$) before normalizing to prevent a single viral video from dominating.
- Binary flags (wins, playoff races) scale directly to 0 or 100.
- Edge case: if every value is identical, the variable returns 50 (no differentiation possible).

Aggregation. The weekly visibility score is the weighted sum of the normalized variables. The season total is the sum of the 36 weekly scores (theoretical maximum $100 \times 36 = 3,600$). Efficiency is the season total divided by the estimated cost in millions of dollars.

1.4 Validation

The model was validated with directional checks (do the rankings match expected patterns), event checks (do wins and playoff races spike), and anomaly detection. Every check passed.

Check	Result	Expected	Pass
Finish position vs. total visibility	$r = -0.93$	Negative (better finish, higher score)	Yes
Wins vs. total visibility	$r = +0.76$	Positive	Yes

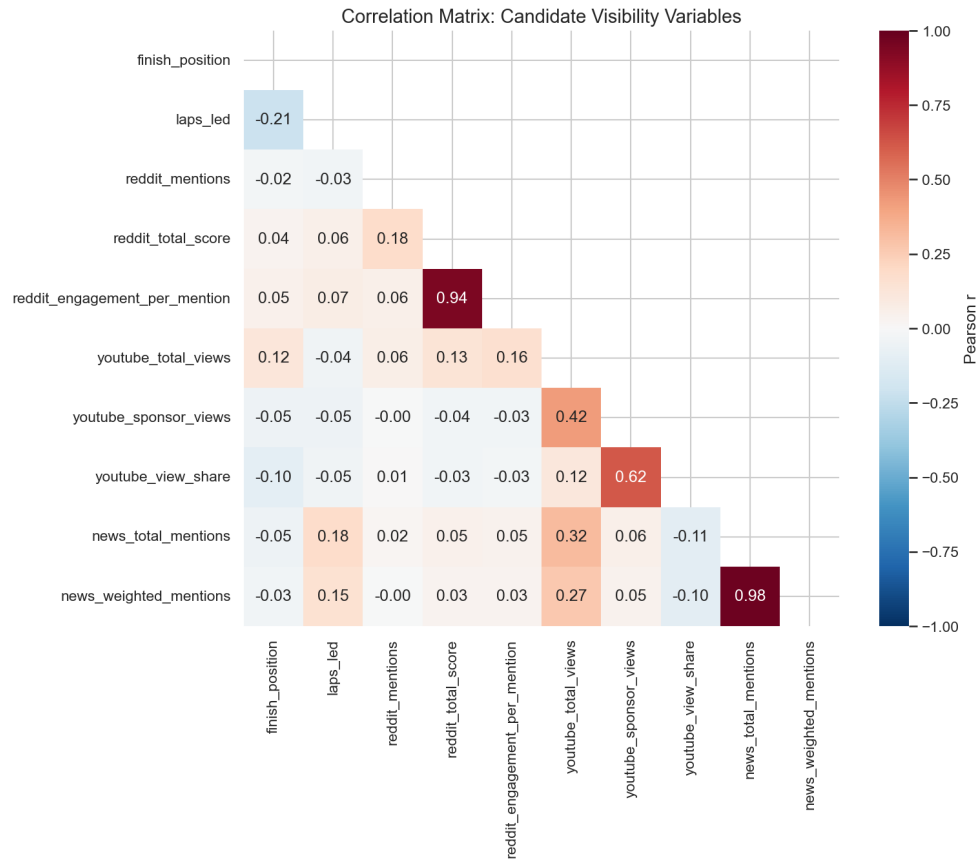
Check	Result	Expected	Pass
Laps led vs. total visibility	$r = +0.70$	Positive	Yes
Win vs. non-win weekly score	1.9x	1.5-3x	Yes
Top-5 vs. bottom-20 weekly score	2.5x	Above 1.5x	Yes
Anomalies (scores over 100, impossible values)	None	None	Yes

A weight-sensitivity test across four configurations (Section 2.4) moved the ranking at most one position, confirming the model is stable.

2. Findings

2.1 Exploratory Analysis

Before building the model we examined how race performance relates to the exposure channels. Contrary to the common assumption that better results drive more buzz, the performance-to-visibility relationship is **weak** in this dataset.



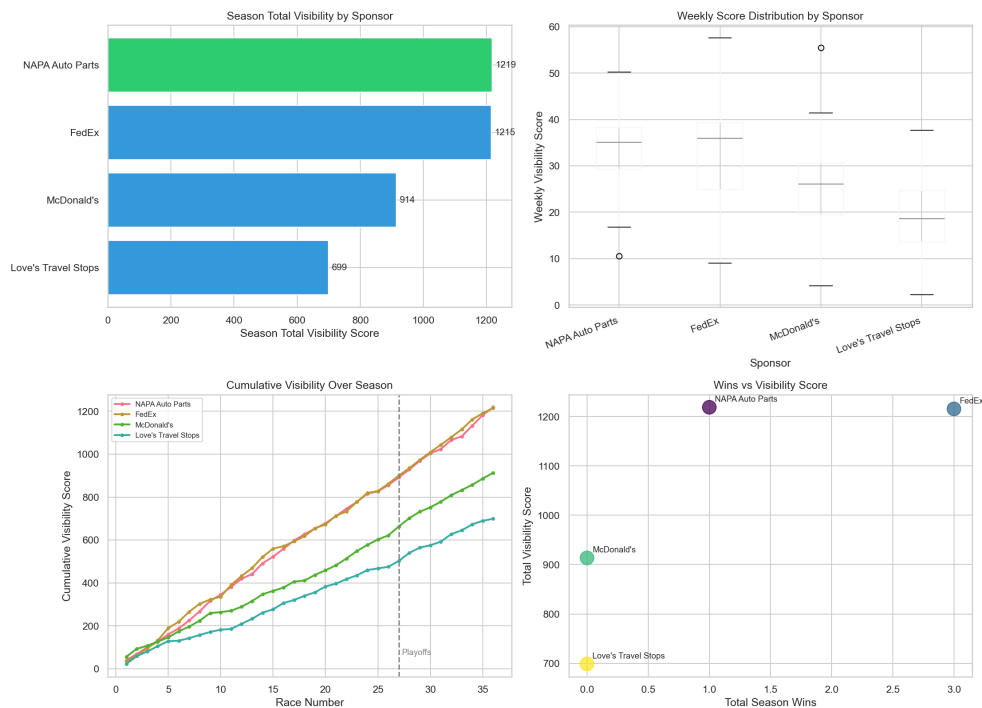
Correlation among candidate variables. Multicollinearity (news_total vs news_weighted $r = 0.98$; reddit_total vs engagement $r = 0.94$) guided which metrics entered the score.

Relationship	Correlation (r)	p-value	Significant?
Laps led vs. news mentions	+0.181	0.030	Yes
Finish position vs. YouTube views	+0.124	0.139	No
Finish position vs. Reddit mentions	-0.024	0.773	No

2.2 Visibility Rankings

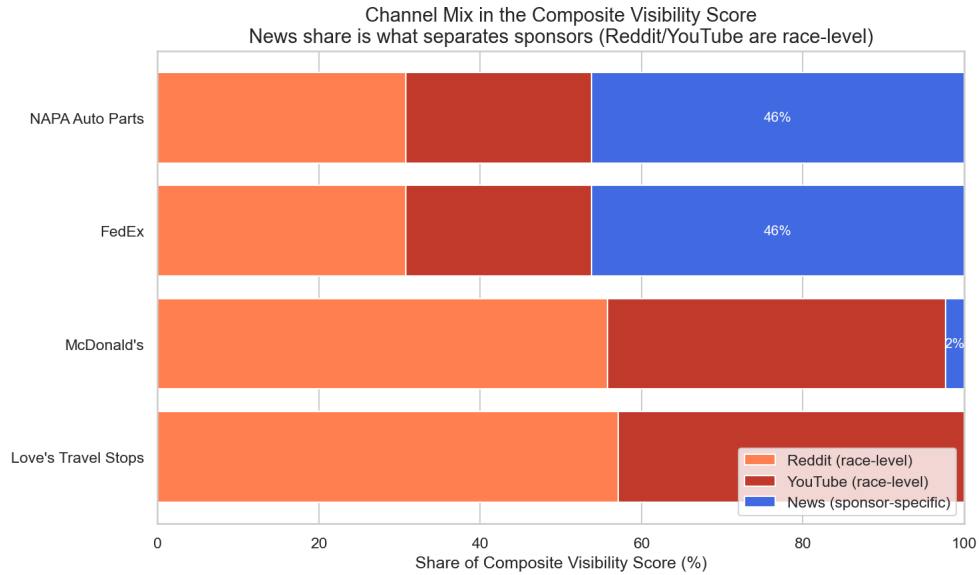
Rank	Sponsor	Total Visibility	Key Driver
1	NAPA Auto Parts	1,219	Consistent top-10 finishes (most, 19); steady exposure
2	FedEx	1,215	Most wins (3) and laps led; boom-or-bust superspeedway peaks
3	McDonald's	914	Mid-pack performance
4	Love's Travel Stops	699	Back-of-field running; lowest news coverage

NAPA and FedEx are effectively tied at the top - a 3-point gap on a roughly 1,200-point scale. The dashboard below shows season totals, the spread of weekly scores, cumulative accumulation across the season, and the relationship between wins and total visibility.



Season totals, weekly score distributions, cumulative trends, and wins vs. visibility.

What drove each score. Because Reddit and total YouTube are race-level, the sponsor-level differences in the composite come mainly from news coverage and on-track performance. The channel mix below shows how large the news contribution is for the top two sponsors relative to the others.



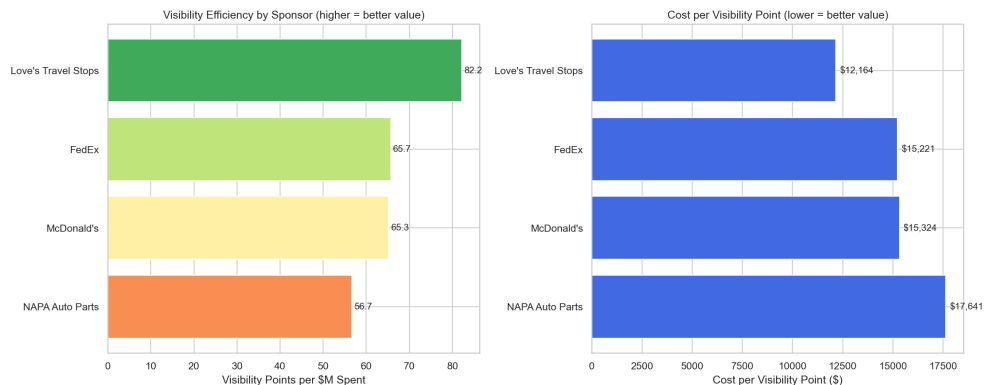
Channel mix within the composite score. The news slice is what separates sponsors; Reddit and YouTube are race-level and identical.

2.3 Efficiency Analysis

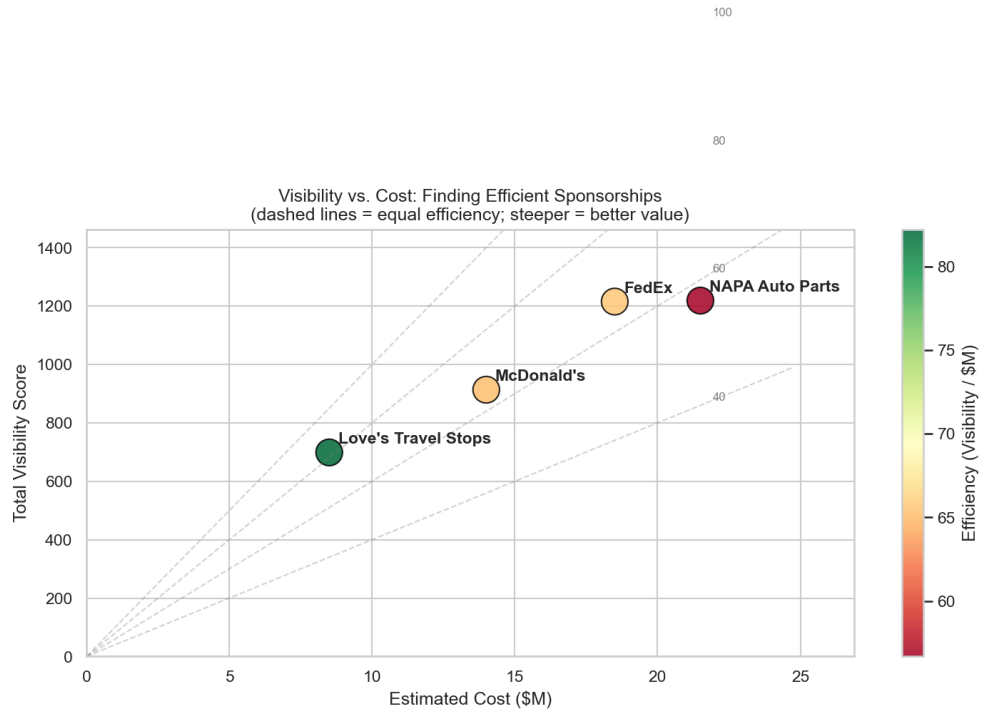
Adding the cost dimension changes the picture entirely.

Rank	Sponsor	Efficiency (pts/\$M)	Est. Cost	Visibility Rank
1	Love's Travel Stops	82.2	\$8.5M	4
2	FedEx	65.7	\$18.5M	2
3	McDonald's	65.3	\$14.0M	3
4	NAPA Auto Parts	56.7	\$21.5M	1

Key insight. The efficiency ranking inverts the visibility ranking. Love's, last in raw exposure, is first in value per dollar (a cheaper Front Row deal); NAPA, first in exposure, is last in efficiency (a premium Hendrick deal). FedEx is the only sponsor top-2 on both dimensions.



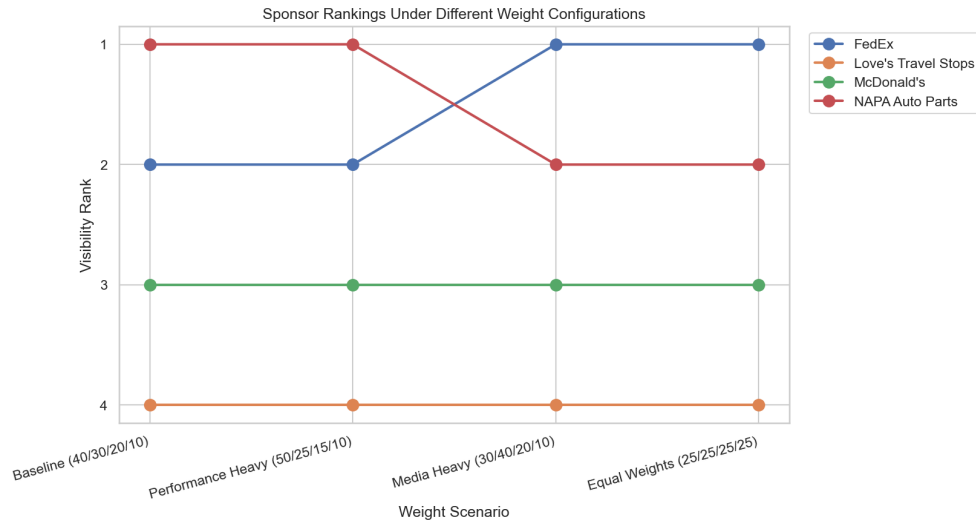
Efficiency (visibility per dollar, left) and cost per visibility point (right) by sponsor.



Visibility vs. cost. Dashed lines are equal-efficiency isolines; a steeper position means better value.

2.4 Sensitivity Analysis

Because both the weights and the costs are judgment calls, we tested how sensitive the rankings are to each. Weights were varied across four configurations (baseline 40/30/20/10, performance-heavy, media-heavy, equal). The ranking is **robust**: the maximum movement is one position, and only the top two (NAPA and FedEx) swap. McDonald's (#3) and Love's (#4) never move. Efficiency ranks were also stable across low, mid, and high cost estimates.



Sponsor ranks under different weight configurations. Only the top two swap; the rest are fixed.

3. Recommendations

3.1 Primary Recommendation: FedEx

FedEx represents the best-balanced sponsorship opportunity for NY Racing.

- **Only sponsor top-2 on both dimensions** (visibility #2, efficiency #2), minimizing the trade-off between exposure and value.
- **Best on-track performance:** 3 wins and 12 top-5 finishes, the most wins of any sponsor.
- **Near-max exposure at a lower price:** 99.7% of the top sponsor's visibility at about 14% lower cost (\$18.5M vs \$21.5M).

Key metrics: visibility 1,215 (rank #2), efficiency 65.7 pts/\$M (rank #2), estimated cost \$18.5M.

Risks: higher weekly volatility (boom-or-bust superspeedway results) and a top-tier price. Mitigation: frame expectations around season totals, and use the efficiency story to justify the spend.

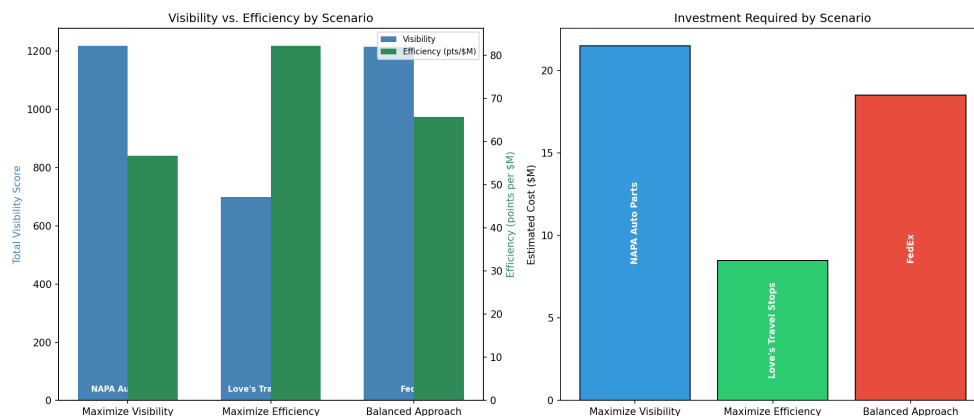
3.2 Alternative: Love's Travel Stops (Value)

Best for teams prioritizing ROI. Love's is #1 in efficiency (82.2 pts/\$M) at the lowest cost (\$8.5M), with low week-to-week volatility. Trade-off: the lowest absolute visibility (rank #4). Choose it when budget is constrained or an ROI justification is critical, and position it as an efficient entry point rather than a headline play.

3.3 Strategic Scenarios

The choice ultimately depends on NY Racing's priorities. Three distinct strategies each point to a different sponsor.

Scenario	Recommended Sponsor	Visibility	Efficiency	Cost	Best For
A. Maximize Visibility	NAPA Auto Parts	1,219 (#1)	56.7 (#4)	\$21.5M	Brand awareness
B. Maximize Efficiency	Love's Travel Stops	699 (#4)	82.2 (#1)	\$8.5M	ROI / budget
C. Balanced (recommended)	FedEx	1,215 (#2)	65.7 (#2)	\$18.5M	Mixed priorities



Visibility vs. efficiency (left) and required investment (right) for each strategy.

3.4 Implementation Guidance

1. **Immediate:** confirm the actual FedEx deal cost and contract expiry (the negotiation window).
2. **Short-term (1-2 weeks):** assess brand alignment between FedEx and NY Racing's target audience.
3. **Medium-term (1 month):** validate the recommendation with multi-season data before a final commitment, and evaluate activation value (hospitality, B2B) that this analysis does not capture.

3.5 Reusable Decision Framework

This methodology can be reapplied to evaluate future sponsorship opportunities.

Step	Action	Deliverable
1	Collect race performance data	Race results dataset

Step	Action	Deliverable
2	Gather social and media exposure	Social + media data
3	Calculate the normalized composite visibility score	Scored dataset
4	Research cost estimates	Cost range table
5	Compute efficiency (visibility per dollar)	Efficiency rankings
6	Run weight and cost sensitivity tests	Stability assessment
7	Make a recommendation with documented rationale	Decision document

4. Limitations

Transparency about what the analysis cannot tell you is what separates a credible study from a marketing claim. Five limitations are material.

#	Limitation	Impact	Mitigation
1	Cost estimates are confidential approximations	Efficiency ranks could shift if actual costs differ	Tested low / mid / high cost estimates; conclusions held
2	Reddit is a Google Trends proxy and, with total YouTube views, is race-level	Those channels do not differentiate sponsors in the ranking	News (sponsor-specific) carries the sponsor-level signal and gets the highest weight
3	Sponsor-specific YouTube views are non-zero in only 4 of 144 rows	The 10% YouTube weight contributes little in practice	Flagged for re-collection at the sponsor level
4	Single 2024 season	May not generalize to future seasons	Framed conclusions as 2024-specific; recommended multi-season validation
5	Activation (hospitality, B2B) not measured	Captures visibility, not total sponsorship ROI	Scope stated clearly as visibility, not full ROI

What this analysis CAN tell you: relative visibility rankings of the analyzed sponsors, which sponsors deliver the most visibility per estimated dollar, how race performance relates to visibility, and which events generate the most exposure.

What it CANNOT tell you: whether a sponsorship is a good business decision overall, the actual (confidential) sponsorship costs, whether past visibility predicts future results, or how visibility converts to sales.

Appendices

Appendix A: Full Data Tables

A.1 Combined Rankings

Sponsor	Total Visibility	Vis Rank	Efficiency (pts/\$M)	Eff Rank	Est. Cost
NAPA Auto Parts	1219	1	56.7	4	\$21.5M
FedEx	1215	2	65.7	2	\$18.5M
McDonald's	914	3	65.3	3	\$14.0M
Love's Travel Stops	699	4	82.2	1	\$8.5M

A.2 Descriptive Statistics by Sponsor

Sponsor	Mean Finish	Median	Std	Best	Worst	Laps Led	Wins	Top-5 Rate
NAPA Auto Parts	11.7	10	8.5	1	36	431	1	31%
FedEx	13.9	10	11.6	1	38	943	3	33%
McDonald's	15.3	14	9.6	3	36	139	0	17%
Love's Travel Stops	21.3	22	10.4	2	38	256	0	6%

A.3 Cost Estimates (benchmark-based; confidential deals)

Sponsor	Team	Tier	Low	Mid	High	Confidence
NAPA Auto Parts	Hendrick Motorsports	top	\$18M	\$21.5M	\$25M	medium
FedEx	Joe Gibbs Racing	top	\$15M	\$18.5M	\$22M	medium-high

Sponsor	Team	Tier	Low	Mid	High	Confidence
McDonald's	23XI Racing	mid-to-top	\$10M	\$14.0M	\$18M	medium
Love's Travel Stops	Front Row Motorsports	mid	\$5M	\$8.5M	\$12M	low-medium

A.4 Multi-Criteria Evaluation (0-100 scores)

Recommendation score = 0.4 visibility + 0.4 efficiency + 0.2 consistency.

Sponsor	Visibility Score	Efficiency Score	Consistency Score	Rec Score
NAPA Auto Parts	100	25	93	68.5
FedEx	75	75	0	60.0
McDonald's	50	50	53	50.5
Love's Travel Stops	25	100	100	70.0

Appendix B: Methodology Details

B.1 Scoring Configuration

Variable	Category	Weight in Category	Effective Weight	Direction	Transform
finish_position	Race Performance	70%	28%	inverse	invert_position
laps_led	Race Performance	30%	12%	direct	normalize
news_weighted_mentions	Media Coverage	100%	30%	direct	normalize
reddit_mentions	Social Engagement	50%	10%	direct	normalize
youtube_sponsor_views	Social Engagement	50%	10%	direct	log_normalize
is_win	Special Events	60%	6%	direct	binary
is_playoff	Special Events	40%	4%	direct	binary

B.2 Variable Selection Decision

Seven variables were selected from ten candidates on three tests (relevance, independence, reliability). Five were excluded.

Selected: finish_position, laps_led (performance); news_weighted_mentions (media); reddit_mentions, youtube_sponsor_views (social); is_win, is_playoff (special events).

Excluded and why: reddit_total_score and reddit_engagement_per_mention (sparse, 9 of 144 rows; redundant, $r = 0.94$); youtube_total_views (race-level, not sponsor-specific); youtube_view_share (derivative and sparse); news_total_mentions (redundant with weighted news, $r = 0.98$).

B.3 Weighting Rationale

Category weights (Race Performance 40%, Media 30%, Social 20%, Special Events 10%) come from industry sponsorship-ROI frameworks (on-field performance 30-40%, media 25-35%, social 20-30%, activation 10-20%), with the unmeasurable activation weight redistributed. The split was then refined by the exploratory analysis: news receives the highest single variable weight (30%) because it was the only sponsor-varying channel that produced a statistically significant link to performance. Finish position is weighted highest within performance on industry grounds, with the open caveat (Section 4) that its observed correlation with exposure was weak in this dataset.

Appendix C: Code Repository Guide

All analysis is reproducible in the project repository.

File / Folder	Purpose
code/clean_race_data.ipynb, code/data_merge_master.ipynb	Data cleaning and merge (Project 2)
code/correlation_analysis.ipynb	Correlation analysis (Project 3.1)
code/visualization_analysis.ipynb	EDA and visualizations (Project 3.2)
code/scoring_methodology.ipynb	Variable selection, weighting, normalization design (Project 4.1)
code/visibility_scoring_model.ipynb	Scoring model implementation and validation (Project 4.2)
code/roi_efficiency_analysis.ipynb	Cost research and efficiency analysis (Project 4.3)
code/strategy_recommendations.ipynb	Recommendation briefs and scenarios (Project 5.1)
code/final_report.ipynb	This report assembly (Project 5.2)
outputs/project4/, outputs/project5/	Generated results, tables, and charts

End of report. Prepared by Aylin Yasgul for NY Racing. Based on 2024 season data.
